INCH-POUND

MIL-C-48397A(AR)
AMENDMENT 4
3 July 1996
SUPERSEDING
AMENDMENT 3
23 August 1994

# MILITARY SPECIFICATION

## CARTRIDGE, IGNITION, M752A1
## LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-C-48397A (AR), dated 22 oCTOBER 1986, and is approved for use by the U.S. Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, Delete: “M853”, and substitute: “M853A1”.

PAGE 4

3.5 Delete in its entirety and substitute:

“3.5 Leak tests. Each body assembly, after the primer is loaded and the lacquer cured, and each loaded ignition cartridge, shall satisfactorily pass the leak tests specified in 4.5.4.”

3.6.2, Delete: “220”, and substitute: “220”.

Delete : “200”, and substitute: “200”.

3.7, Delete: “210”, and substitute: “190”.



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PAGE 6

Table I, under "Body Assembly (dwg. 12630551)", delete:

"Push-out test . . . 10 . . . 100% . . . 3.2 . . . 4.4.3.1/4.5.1".

PAGE 9

4.4.2.2, Major 102: Delete in its entirety.

Major 103: Delete: "Gage", and substitute: "Visual".

PAGE 10

4.4.2.3, Major 101, 102, 103, 105 and 106: Delete in their entirety and substitute:

"101 Primer missing or inverted . . . 100% . . . 3.2 . . . visual

102 Primer above flush or more than .015 in. below flush . . . 0.4% . . . 3.2 . . . Gage

103 Lacquer seal missing or inadequate . . . ( ) . . . 100% . . . 3.2 . . . Gage/4.5.4.1

105 Staking process inadequate . . 2/ . . . ( ) . . . 3.2 . . . Gage/4.4.3.1

106 Primer bulged after staking . . . 100% . . . 3.2 . . . Visual".

Major 104, Delete: "lacquer", and substitute: "any adhesive or sealant".

Add Major 107 as follows:

"107 One or more stakes missing or not full . . . 1/ . . . 3.2 . . . Visual"

Note "(a)", Delete in its entirety and substitute:

"1/ Inspect the tool or one body at the start of each day's production and each hour thereafter. A missing or broken tooth shall be cause for rejection of all parts staked since the last successful inspection.

2/ See 4.4.3.1".



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4.4.2.4, Delete: drawing number “9549177”, and substitute: “7549177”.

Major 106, Delete in its entirety.

PAGE 12

4.4.2.5, Add Critical 5 as follows:

“5 Firing plug above flush with head surface . . . 100% . . . 3.2 . . .Gage” .

Majors 101, 102, 103 and 104, Delete in their entirety

Major 106, Delete: “Gage”, and substitute: “Visual”.

Major 107, Change “Pellet missing” to “Pellet missing or broken c/”

Add Note c/ as follows:

“c/ Minor chipping of edges acceptable, no chunks or excessive chipping.”

PAGE 13

4.4.2.6, Add Major 103 as follows:

“103 Environmental conditions inadequate . . . d/ . . . 3.2 . . . Gage”.

Add Note d/ as follows:

“a/: Temperature and humidity shall be checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected.”



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PAGE 14

4.4.2.7, Major 101, Delete examination callout in its entirety, and substitute:

"Channel not completely filled with RTV sealant".

Major 104, Delete in its entirety.

Major 105, Delete in its entirety and substitute:

"105 True position of outside diameter with major diameter of head thread . . . 0.40% . . . 3.2 . . . Gage".

Major 107, Delete "4.5.4 and substitute "4.5.4.2".

Add Majors 110, 111 as follows:

"110 Environmental conditions inadequate . . . b/ . . . 3.2 ...Gage

111 Presence of excessive sealant . . . 0.40% . . . 3.2 ...Visual".

Add Note b/ as follows:

"b/ Temperature and humidity shall be checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected."

PAGE 20

4.4.3.1, Delete in its entirety and substitute:

"4.4.3.1 Staking process (primer security). Each hour, and on the last units staked before a scheduled staking tool change, five (5) primed bodies shall be inspected before and after staking, using any method that can verify that the primer cups have been pushed down as a result of the staking operation. If none of the primer cups have been pushed down, all units loaded since the last successful inspection shall be rejected. If one or more primer cups have been pushed down by any amount, the process may continue unchanged at the discretion of the contractor."



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PAGE 21

4.4.3.4.1, Delete: “0.025”, and substitute: “0.25”.

4.4.3.4.1.1, Delete: “0.040”, and substitute: “0.40”.

PAGE 22

4.5.1, Delete in its entirety.

4.5.2, Delete the text in its entirety and substitute:

“The moisture content of the black powder pellets shall be determined in accordance with the procedures specified in MIL-P-223. A sample of 25 black powder pellets shall be randomly selected from each production shift. If any pellet fails to comply with the requirement cited in paragraph 3.3 herein, all pellets in the production shift represented by this sample shall be rejected. If loading with propellant has begun, ignition cartridges loaded with the non–conforming black powder pellets shall be rejected.”

PAGE 23

4.5.3.2, Delete sentence 2 in its entirety and substitute:

“The firing pin of the test fixture shall be adjusted so that it impacts the firing plug at 0.025 inch off-center. ”

PAGE 24

4.5.4, Delete in its entirety and substitute:

“4.5.4 Leak tests.

4.5.4.1 Body assembly. After the lacquer has cured, each primed body assembly shall be tested for leakage around the primer. The pressure differential and duration of the test shall be as specified for the leak test of the fully assembled ignition cartridge (see drawing 9278622). Any body assembly which fails the test shall be classed defective and removed from the lot.


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4.5.4.2 Ignition cartridge. Each fully assembled ignition cartridge shall be tested in accordance with drawing 9278622. Any ignition cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot."

4.5.6, Delete: "8.9 ± 0.02 lbs for M853", and substitute: "10.35 ± 0.02 lbs for M819".

4.5.7, Delete: "8.9 ±", and substitute: "10.35 ±".

Delete: "M853", and substitute: "M819".

Delete: "Develoment", and substitute "Development".

PAGE 25

6.1, Delete: "M853", and substitute: "M853A1"

6.3, Last sentence, delete: ". . . AMSMC-QAT-I, Dover, NJ 07801-5000" and substitute:

"AMSTA-AR-QAA-C, Picatinny Arsenal, NJ 07806–5000."

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